Failure of Antisymmetry and of the Cyclic Jacobi Identity for a Specified Finite-Fourier Tethered Operation

Companion paper II (machine-checked, finite-Fourier model)

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Abstract

In a finite-Fourier model of vector fields on $\mathbb{T}^3$ we consider the vector-valued tethered bracket $\{a,b\} = [a,b] + T(a,b)$ with tether axis $u_Y = (\sin y, 0, 0)$. Take the divergence-free field $a_Z = (0, 0, \sin x)$. If $\kappa w \neq 0$, the Jacobiator $J(a_Z, a_Z, a_Z)$ is nonzero: its z-component at mode $e^{\mathrm{i}(x+y)}$ equals $\frac{3\mathrm{i}}{8}\kappa w$. A separate finite-dimensional witness, with $u = e_1$ and the cross-product bracket on $\mathbb{R}^3$, gives a nonzero Jacobiator with value $(\kappa w)e_2$. Both results are formalized in Lean 4/Mathlib. Because T is symmetric, the operation is not antisymmetric, and both witnesses repeat arguments. This is a valid algebraic failure of the cyclic Jacobi identity for this specified operation. It is *not* a failure of the Lie part (which satisfies Jacobi), *not* a Poisson-bracket failure, and *not* a geometric refutation. Companion paper I treats the restricted shear class.

1 The finite-Fourier field model (exact Lean definitions)

All definitions are in `FTVerified/May/FieldJacobi.lean` (namespace `Frohman.Tether.FieldJacobi`), unchanged since commit `26f16da`. For $k \in \mathbb{Z}^3$ write $k_{\mathbb{C}} \in \mathbb{C}^3$ for its image (`kC`, `BiotSavart.lean:46`); $v \cdot w = \sum_j v_j w_j$ is the *bilinear* (non-Hermitian) dot product on $\mathbb{C}^3$. Components are indexed $0, 1, 2$ $(= x, y, z)$.

Definition 1 (`Fld`, line 48)**.** `Fld` is the space of finitely supported maps $a : \mathbb{Z}^3 \to \mathbb{C}^3$, $k \mapsto \hat{a}_k$; we read a as the trigonometric polynomial $\sum_k \hat{a}_k e^{\mathrm{i}k\cdot x}$ on $\mathbb{T}^3$. No reality condition is imposed; scalars are complex.

Definition 2 (`bvec`, `br`, lines 51–56)**.** For modes p, q and $v, w \in \mathbb{C}^3$, $\mathrm{bvec}(p, q, v, w) = \mathrm{i}\big((v \cdot q)\, w - (w \cdot p)\, v\big)$, and $[a,b] = \sum_{p,q} \mathrm{bvec}(p, q, \hat{a}_p, \hat{b}_q)\, \delta_{p+q}$, the Fourier form of $(a \cdot \nabla)b - (b \cdot \nabla)a$.

Definition 3 (`pr`, line 59)**.** $\langle a, b\rangle = \sum_p \hat{a}_p \cdot \hat{b}_{-p}$ (complex-bilinear L^2 pairing $(2\pi)^{-3}\int a \cdot b$).

Definition 4 (`piF`, `tetF`, `brT`, `jacT`; lines 108, 111, 362, 364)**.** For $u, x \in$ `Fld` and $\kappa, w \in \mathbb{C}$: $\Pi_u x = x - \frac{\langle u,x\rangle}{\langle u,u\rangle} u$ (Lean division, so $\Pi_u x = x$ if $\langle u, u\rangle = 0$); $T(a,b) = -(\kappa w)\langle \Pi_u a, \Pi_u b\rangle\, u$; $\{a,b\} = [a,b] + T(a,b)$; $J(a,b,c) = \{\{a,b\},c\} + \{\{b,c\},a\} + \{\{c,a\},b\}$. Here κ, w are *arbitrary complex parameters*: in this file w is not tied to any field (the docstring of `Basic.lean` reads w as $\int |\omega|^2$, but nothing in `FieldJacobi.lean` enforces that).

Definition 5 (`mixedF`, `mixedSumF`; lines 114, 116)**.** $M(a,b,c) = [T(a,b),c] + T([a,b],c)$ and $S(a,b,c) = M(a,b,c) + M(b,c,a) + M(c,a,b)$.

Definition 6 (`DivergenceFree`, line 121; `XOnly`, line 183)**.** a is *divergence-free* if $k_{\mathbb{C}} \cdot \hat{a}_k = 0$ for all k; a is *XOnly* if $(\hat{a}_k)_1 = (\hat{a}_k)_2 = 0$ for all k.

Definition 7 (explicit fields, lines 224–234)**.** $k_x = (1,0,0)$, $k_y = (0,1,0)$. $u_Y = \delta_{k_y}(-\frac{\mathrm{i}}{2},0,0) + \delta_{-k_y}(\frac{\mathrm{i}}{2},0,0)$, i.e. $(\sin y, 0, 0)$; $a_Z = \delta_{k_x}(0,0,-\frac{\mathrm{i}}{2}) + \delta_{-k_x}(0,0,\frac{\mathrm{i}}{2})$, i.e. $(0,0,\sin x)$.

Lemma 8 (Jacobiator reduction, `jacT_eq_mixedSum`, line 384)**.** *If $\langle u,u\rangle \neq 0$ then $J(a,b,c) = S(a,b,c)$ for all a,b,c and all κ, w.*

Proof. T is additive in its first slot (`tetF_add_left`) and $[\cdot,\cdot]$ is additive in each slot, so $\{\{a,b\},c\} = [[a,b],c] + M(a,b,c) + T(T(a,b),c)$. Since $T(a,b) = s\,u$ and $\Pi_u(su) = su - \frac{s\langle u,u\rangle}{\langle u,u\rangle}u = 0$ when $\langle u,u\rangle \neq 0$ (`piF_smul_self`), $T(T(a,b),c) = 0$ (`tetF_tetF`). Summing cyclically, $J = \mathrm{jacF}(a,b,c) + S(a,b,c)$, where jacF is the Jacobiator of $[\cdot,\cdot]$. Finally $\mathrm{jacF} \equiv 0$ (`jacF_eq_zero`, line 164): by trilinearity it suffices to check single modes, where all three terms sit at mode $p+q+r$ and the identity is a polynomial identity in the coordinates of v,w,z,p,q,r (closed by `ring`); it is the Fourier form of Jacobi for the Lie bracket of vector fields. □

Lemma 9 (`pr_uY_uY`, line 396)**.** $\langle u_Y, u_Y\rangle = \frac{1}{2}$*; in particular* $\neq 0$.

Proof. $\langle u_Y,u_Y\rangle = (-\frac{\mathrm{i}}{2})(\frac{\mathrm{i}}{2}) + (\frac{\mathrm{i}}{2})(-\frac{\mathrm{i}}{2}) = \frac{1}{4} + \frac{1}{4}$. □

2 Main theorem: the field-level counterexample

Lemma 10 (diagonal reduction, `mixedSumF_diag`, line 257)**.** *For all u,a:* $S(a,a,a) = 3\big(-(\kappa w)\langle \Pi_u a, \Pi_u a\rangle\big)\,[u,a]$.

Proof. $[a,a] = 0$ by antisymmetry (`br_antisymm`, `br_self`), so $T([a,a],a) = T(0,a) = 0$. Each of the three equal terms is $M(a,a,a) = [s\,u,a] = s[u,a]$ with $s = -(\kappa w)\langle \Pi_u a, \Pi_u a\rangle$ (`br_smul_left`). □

Lemma 11 (explicit values; lines 265, 269, 273, 299, 295)**.** *(a) $\langle u_Y, a_Z\rangle = 0$, hence $\Pi_{u_Y} a_Z = a_Z$. (b) $\langle a_Z, a_Z\rangle = \frac{1}{2}$. (c) $\big([u_Y,a_Z]\big)_{k_y+k_x,\,2} = -\frac{\mathrm{i}}{4}$. (d) a_Z and u_Y are divergence-free.*

Proof. (a) Every term pairs an x-directed vector with a z-directed one, so it is 0. (b) $(-\frac{\mathrm{i}}{2})(\frac{\mathrm{i}}{2}) + (\frac{\mathrm{i}}{2})(-\frac{\mathrm{i}}{2}) = \frac{1}{2}$. (c) The mode $k_x + k_y = (1,1,0)$ arises only from $p = k_y$, $v = (-\frac{\mathrm{i}}{2},0,0)$ and $q = k_x$, $w = (0,0,-\frac{\mathrm{i}}{2})$. Here $v \cdot q = -\frac{\mathrm{i}}{2}$ and $w \cdot p = 0$, so $\mathrm{bvec} = \mathrm{i}(-\frac{\mathrm{i}}{2})w = \frac{1}{2}w$, whose z-component is $-\frac{\mathrm{i}}{4}$. (d) The modes of a_Z are $\pm k_x$, and $(\pm 1,0,0)\cdot(0,0,*) = 0$. The modes of u_Y are $\pm k_y$, and $(0,\pm 1,0)\cdot(*,0,0) = 0$. □

Theorem 12 (`shearY_transverse_coeff`, line 347)**.** *For all $\kappa, w \in \mathbb{C}$: $S(a_Z,a_Z,a_Z)_{k_x+k_y,\,2} = \frac{3\mathrm{i}}{8}\kappa w$ at $u = u_Y$.*

Proof. By Lemmas 10 and 11: $3 \cdot (-\kappa w \cdot \frac{1}{2}) \cdot (-\frac{\mathrm{i}}{4}) = \frac{3\mathrm{i}}{8}\kappa w$. □

Theorem 13 (`shearY_jacobi_fails_transverse`, line 406)**.** *If $\kappa w \neq 0$ then $J(a_Z,a_Z,a_Z) \neq 0$ for the tethered bracket at $u = u_Y$.*

Proof. $\langle u_Y,u_Y\rangle = \frac{1}{2} \neq 0$, so $J = S$ by Lemma 8. If $S(a_Z,a_Z,a_Z) = 0$, Theorem 12 would give $\frac{3\mathrm{i}}{8}\kappa w = 0$ and hence $\kappa w = 0$, which is a contradiction (`shearY_transverse_ne_zero`). □

Remark 14 (Precise strength of the statement). (i) The Lean theorem's hypothesis is only $\kappa w \neq 0$. Divergence-freeness of a_Z and u_Y is a *separate* fact (`divFree_aZ`, `divFree_uY`) and is not a hypothesis of the theorem. (ii) *Antisymmetry fails.* T is symmetric, so $\{a, a\} = T(a, a)$ can be nonzero. Here $\{a_Z, a_Z\} = -\frac{\kappa w}{2} u_Y$, which is nonzero when $\kappa w \neq 0$. This value is the Lean theorem `brT_uY_aZ_aZ` (from `br_self`, `pr_uY_aZ`, `pr_aZ_aZ`: $\{a_Z, a_Z\} = 0 + T(a_Z, a_Z) = -(\kappa w) \cdot \frac{1}{2} \cdot u_Y$), and `brT_not_antisymm` proves $\{a_Z, a_Z\} \neq -\{a_Z, a_Z\}$ when $\kappa w \neq 0$: if $v = -v$ then $2v = 0$, and $2 \cdot (-\frac{\kappa w}{2}) = -\kappa w \neq 0$ while $u_Y \neq 0$. So the operation is not antisymmetric. The antisymmetric part of $\{\cdot, \cdot\}$ is the field Lie bracket, which satisfies Jacobi (`jacF_eq_zero`). On the diagonal, $J(a, a, a) = 3\{\{a, a\}, a\}$, and it is nonzero because of the symmetric part. (ii′) *Both witnesses repeat arguments:* the field witness is (a_Z, a_Z, a_Z) with complex κ, w, and the $\mathbb{R}^3$ witness is (e_2, e_2, e_3) with real κ, w. No witness with three distinct arguments is proved. (ii″) The result is therefore a valid algebraic failure of the cyclic Jacobi identity for this specified operation. It is *not* a failure of the Lie part, *not* a Poisson-bracket failure, and *not* a geometric refutation. (ii‴) *Non-vacuity* (`paperII_nonvacuous`): at $\kappa = w = 1$ all statements hold with nonzero values, so the hypothesis $\kappa w \neq 0$ is satisfiable.
(iii) κ, w are free complex parameters. In particular w is not computed from a_Z or u_Y. (iv) A related witness in the file, `shearX_jacobi_fails` (line 401), uses $u_X = (\sin x, 0, 0)$. That field is *not* divergence-free (`not_divFree_uX`), so it is not used here.

3 Vector-level witness

`FTVerified/May/JacobiWitness.lean` works in the abstract framework of `Jacobi.lean`. Let $H = \mathbb{R}^3$ (Euclidean), $P(a, b) = a \times b$ (`crossBr`, a classical bracket: `crossBr_isClassical`), $u = e_1$, and $T(a, b) = -(\kappa w)\langle \Pi_u a, \Pi_u b\rangle u$ with Π_u the orthogonal projection onto $u^\perp$ and $\kappa, w \in \mathbb{R}$.

Theorem 15 (`mixed_sum_eq`:60, `tethered_jacobiator_ne_zero`:78)**.** *The mixed sum at (e_2, e_2, e_3) equals $(\kappa w)e_2$. Hence, if $\kappa w \neq 0$, the Jacobiator of $P + T$ at (e_2, e_2, e_3) is nonzero.*

Proof. $M(e_2, e_2, e_3) = T(e_2, e_2) \times e_3 + T(e_2 \times e_2, e_3) = -\kappa w\, e_1 \times e_3 + 0 = \kappa w\, e_2$. $M(e_2, e_3, e_2) = T(e_2, e_3) \times e_2 + T(e_1, e_2) = 0$, because $\langle e_2, e_3\rangle = 0$ and $\Pi_u e_1 = 0$. $M(e_3, e_2, e_2) = T(e_3, e_2) \times e_2 + T(-e_1, e_2) = 0$. The Jacobiator equals the mixed sum (`jacobiator_eq_mixed`: the classical terms cancel by Jacobi for $\times$, and the TT terms vanish because $\Pi_u u = 0$). □

This witness is finite-dimensional linear algebra. It has no notion of a velocity field, and divergence-freeness does not enter it. It is *not* a Fourier-field statement, and it should not be read as one.

4 Lean mapping

Statement	Lean name (file:line)	Exact hypotheses
Thm. 13	`FieldJacobi.shearY_jacobi_fails_transverse`:406	{κ w : $\mathbb{C}$}, h : κ * w $\neq$ 0; concludes jacT κ w uY aZ aZ aZ $\neq$ 0
Thm. 12	`FieldJacobi.shearY_transverse_coeff`:347	none (any κ, w)
	`FieldJacobi.shearY_transverse_ne_zero`:352	$\kappa w \neq 0$
Lem. 10	`FieldJacobi.mixedSumF_diag`:257	none
Lem. 11	`pr_uY_aZ`:265, `pr_aZ_aZ`:269, `br_uY_aZ_coeff`:273, `divFree_aZ`:299, `divFree_uY`:295	none
Lem. 8	`FieldJacobi.jacT_eq_mixedSum`:384	pr u u $\neq$ 0
Thm. 15	`JacobiWitness.tethered_jacobiator_ne_zero`:78; `mixed_sum_eq`:60	{κ w : $\mathbb{R}$}, h : κ * w $\neq$ 0
Rem. (ii)	`brT_uY_aZ_aZ`, `brT_not_antisymm` (Companion)	none; $\kappa w \neq 0$ for the second
Rem. (ii‴)	`paperII_nonvacuous` (Companion)	none

5 Provenance, reproduction and axioms

- Full Lean source: `https://github.com/BenFrohman/frohmanian-tether-verified`, code commit `cf04afd5cd9b314703a5c8b196d83ec6b7979e6d` (pull request #2; Apache-2.0). The definitions and the original theorems are in `FTVerified/May/FieldJacobi.lean` and `JacobiWitness.lean`, unchanged since `26f16da`; the new theorems are in `FTVerified/Companion/JacobiCompanion.lean`. Line numbers refer to these files. Geometric comparison repository: `BenFrohman/frohmanian-tether-geometric` at `9e40ae7`.
- Lean toolchain (`lean-toolchain`): `leanprover/lean4:v4.30.0-rc1`.
- Mathlib (`lake-manifest.json`): input rev `v4.30.0-rc1`, locked rev `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`.

One-command rebuild (repository root, at the commit above):

```
lake exe cache get && lake build && python3 scripts/check_jacobi_axioms.py
```

GitHub Actions ran this, together with a comment-aware scan for `sorry`, `admit`, `native_decide` and `axiom` over all 31 project modules, in run `https://github.com/BenFrohman/frohmanian-tether-verified/actions/runs/42923330722` (passed: full library build of 8295 jobs; `PASS: 49 declarations, standard axioms only`). The axiom lines for the declarations used in this paper are (full names are the bracketed namespace prefix `Frohman.Tether.` plus the quoted name):

```
[Frohman.Tether.FieldJacobi]
  'FieldJacobi.jacT_eq_mixedSum'
  'FieldJacobi.mixedSumF_diag'
  'FieldJacobi.pr_uY_aZ'
  'FieldJacobi.pr_aZ_aZ'
  'FieldJacobi.br_uY_aZ_coeff'
```

```
  'FieldJacobi.divFree_aZ'
  'FieldJacobi.divFree_uY'
  'FieldJacobi.shearY_transverse_coeff'
  'FieldJacobi.shearY_transverse_ne_zero'
  'FieldJacobi.shearY_jacobi_fails_transverse'
[Frohman.Tether.JacobiCompanion]
  'JacobiCompanion.brT_uY_aZ_aZ'
  'JacobiCompanion.brT_not_antisymm'
  'JacobiCompanion.paperII_nonvacuous'
[Frohman.Tether.JacobiWitness]
  'JacobiWitness.mixed_sum_eq'
  'JacobiWitness.tethered_jacobiator_ne_zero'
  each: depends on axioms: [propext, Classical.choice, Quot.sound]
```

The checker fails the build if any audited declaration depends on anything beyond `propext`, `Classical.choice` and `Quot.sound`.

6 Comparison with the geometric repository

The repository `frohmanian-tether-geometric` at commit `9e40ae7` formalizes a different object. Line numbers below refer to that commit.

- *Geometric bracket* (`SymplecticTether.lean` 353–381). It is a *scalar* bracket of functionals, $\{F, G\} = \text{ClassicalBracket}(F, G)(\omega) + \text{TetherKernel}(\omega, F, G)$, with

$$\text{TetherKernel}(\omega, F, G) = -\kappa\, \text{Enstrophy}(\omega) \int \left\langle \Pi_{u(\omega)}\, \delta F(x),\ \Pi_{u(\omega)}\, \delta G(x) \right\rangle dx.$$

  The tether axis is state-dependent: $u(\omega)$ is `velocity_from_vorticity` ω, the Biot–Savart velocity (`ArnoldGeometric.lean` 918).

- *Parameters.* The geometric bracket has no free parameter w. Its tether prefactor is $-\kappa \cdot \text{Enstrophy}(\omega)$, with κ fixed at $3/2$ (`kappa_eq_three_halves`, `SymplecticTether.lean` 300–310), and $\text{Enstrophy}(\omega) = \int \|\omega\|^2$ is a state functional (`ArnoldGeometric.lean` 208–213).

- *Projection.* Π_u is the global L^2 Gram–Schmidt projector $v \mapsto v - \frac{\int v \cdot u}{\int \|u\|^2} u$ (`ForMathlib/Projection.lean` 35).

- *Functional derivative.* δF is a chosen Gâteaux representative in the vorticity pairing, and it is 0 if no representative exists (`FunctionalDerivative`, `ArnoldGeometric.lean` 313). For the kinetic energy the representative is the stream function ψ, not the velocity (`SymplecticTether.lean` 366–368).

- *Fourier model (here).* u is *frozen*, $\kappa, w \in \mathbb{C}$ are free, and $\{a, b\}$ is a *vector-valued* operation on velocity-slot fields.

Shared features (stated, not an identification). The geometric operation is also not antisymmetric, because its correction is symmetric. Its diagonal tether is negative semidefinite: $\{F, F\} = T(F, F) = -\kappa \|\omega\|^2 \int \|\Pi_u \delta F\|^2 \le 0$ (`SymplecticTether.lean` 375–380). No geometric Jacobi witness is proved in that repository. Its `tethered_jacobi_identity` (`SymplecticTether.lean` 1187–1196) is conditional on the hypotheses `hident` (the Jacobiator equals an integrated cyclic Lie pairing) and `hvanish` (that integral vanishes); its proof is the rewrite `rw [hident, hvanish]`.

Why nothing transports. Pairing the vector operation here with a state m produces a factor $\langle m, u\rangle$ from the tether term. Even formally, the parameters could therefore be matched only pointwise, where that pairing is nonzero. No identification between the two models is claimed. *Informal remark (not proved):* because u, Enstrophy and $\Pi_{u(\omega)}$ depend on the state, one expects a geometric Jacobiator to contain extra terms involving δE and $\delta\Pi$, which the frozen-u `jacT` lacks. Neither the differentiability of the state-dependent Π nor any geometric Jacobiator formula is proved in either repository. Neither the restricted identity (paper I) nor the counterexample (paper II) is claimed to transport to the geometric setting. This comparison is not a refutation, or a confirmation, of any geometric statement.

7 Audit scope

- The result concerns the *finite-Fourier field model* Fld of complex trigonometric-polynomial vector fields on $\mathbb{T}^3$, with the vector-valued bracket $\{a, b\} = [a, b] + T(a, b)$ defined above. It is not a statement about a Poisson bracket on functionals, about smooth or Sobolev fields, or about Navier–Stokes solutions.
- No identification of this model, or of the shear class, with the geometric reduced coadjoint orbit (or with $u^\perp$ in any geometric sense) has been proved in Lean or here. The link from Fourier coefficients to the `fderiv`-based fields of `Torus.lean` is not formalized for `br`.
- Kernel checking establishes that the stated Lean propositions follow from the stated definitions. It does not establish that the definitions faithfully model the intended geometry, nor any novelty of the results.
- Nothing here bears on any Clay Millennium Problem, and no such claim is made.

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8 Relation to companion paper I

Paper I (*A Restricted Jacobi Identity (Elementary Checked Boundary Case) for the Field-Level Tethered Bracket on the Divergence-Free Shear Class*) proves `shearY_jacobi_holds` (line 410). At the same u_Y, $J(a, b, c) = 0$ whenever a, b, c are XOnly and divergence-free. The field a_Z here is divergence-free but not XOnly, so Theorem 13 shows that the XOnly hypothesis of paper I cannot be dropped: an explicit divergence-free diagonal triple fails. There is no contradiction between the two papers.